Nonlinearity and Risk Control

From Lectures 1–3

Define the forecast $\rightarrow$ test it honestly $\rightarrow$ control fitting sensitivity.

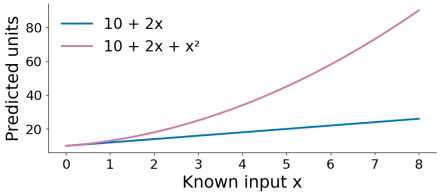
- Today: change the shape of the prediction rule.
- Build a tree, combine trees, evaluate the whole procedure.

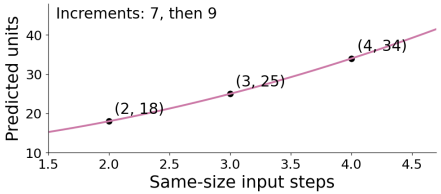
A Constant Change or a Changing Change?

$$\hat{y} = 10 + 2x.$$

A linear response has a constant increment; a nonlinear increment can depend on the input level.

1. At x=2 and 3, linear predictions are 14 and 16.
2. At x=7 and 8, predictions are 24 and 26.
3. Both increments are 2; compare the curved response.





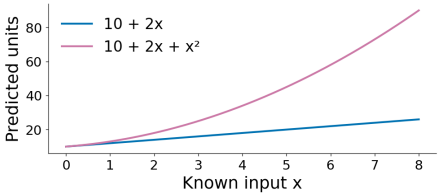
Lecture code 4.1; Separate homework HW4.1. Constructed teaching example.

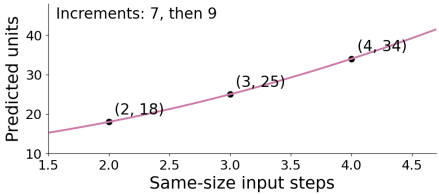
A Squared Input Creates Curvature

$$\hat{y} = 10 + 2x + x^2.$$

Transformation: construct an input from known information. Quadratic: include the square; its degree is two.

1. Square inputs 2, 3, 4 to obtain 4, 9, 16.
2. Add the contributions to obtain 18, 25, 34.
3. Equal input steps now add 7 and 9 predicted units.



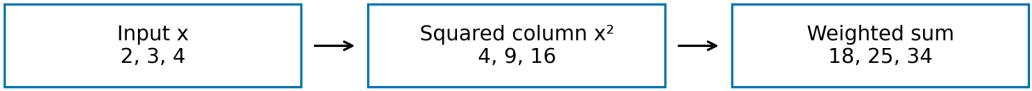


Lecture code 4.1; Separate homework HW4.1. Constructed teaching example.

Linear in Weights Can Still Bend in the Input

Linear in coefficients: weights enter additively even when the response bends in the original input.

1. Keep both columns: x and its computed square.
2. Fit one weight on each column.
3. A weighted sum can bend in the original input.



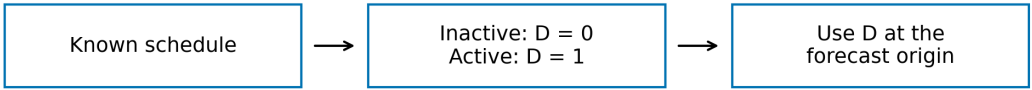
Lecture code 4.1; Separate homework HW4.1. Constructed teaching example.

Threshold, Indicator and Regime

$$D = 1 \text{ if promotion is active;} \qquad D = 0 \text{ otherwise.}$$

Threshold: cutoff. Indicator: one if a condition holds, zero otherwise. Regime: a modeled setting.

1. Read the promotion schedule available at the origin.
2. Encode inactive as 0 and active as 1.
3. Use that value without looking at the future outcome.



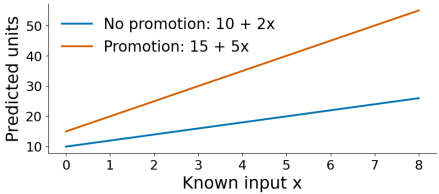
Lecture code 4.2; Separate homework HW4.2. Constructed teaching example.

An Interaction Changes Another Input's Contribution

$$\hat{y} = 10 + 2x + 5D + 3xD.$$

An interaction lets the response to x depend on another input.

1. At x=2, set D=0: prediction 14.
2. At x=2, set D=1: prediction 25.
3. Compare starting levels 10/15 and slopes 2/5.



D = 0: 5D and 3xD disappear

D = 1: starting level 15; slope 5

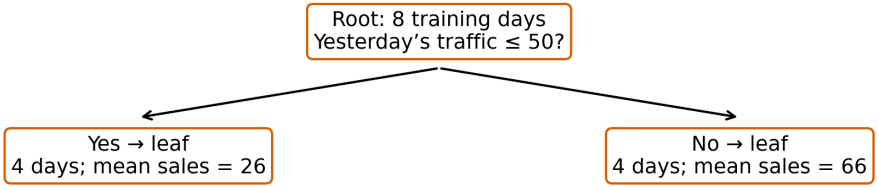
Constructed conditional responses

Lecture code 4.3; Separate homework HW4.3. Constructed teaching example.

A Regression Tree Predicts Through Questions

Root: start node. Split: input question. Branch: answer. Leaf: final numerical prediction group.

1. Start at the root: eight days, traffic cutoff 50.
2. Follow a branch to a leaf mean of 26 or 66.
3. Count split questions along the path to measure depth.



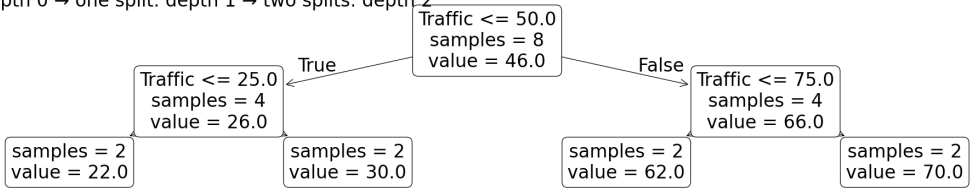
Lecture code 4.4; Separate homework HW4.4. Constructed teaching example.

A Path and Its Depth

Depth counts split questions along a path; the root has depth zero.

1. Start at root depth 0.
2. After one question, reach depth 1; after two, depth 2.
3. Read the fitted leaves 22, 30, 62, 70.

Root depth 0 → one split: depth 1 → two splits: depth 2

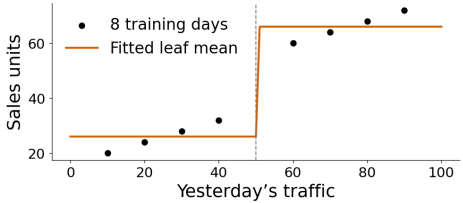


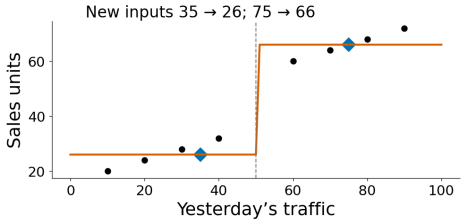
Lecture code 4.4; Separate homework HW4.4. Constructed teaching example.

Compute Two Leaf Predictions

A squared-error regression leaf predicts its training outcome mean.

1. Split the same eight training pairs at traffic 50.
2. Average left sales to 26 and right sales to 66.
3. Route inputs 35/75 to predictions 26/66.



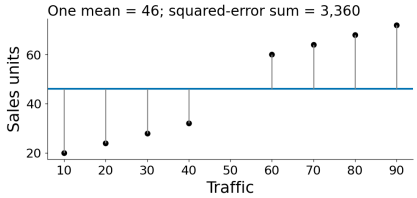


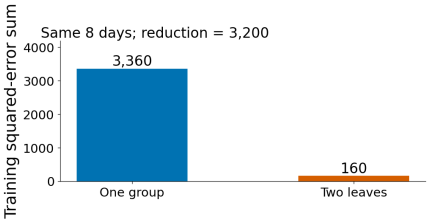
Lecture code 4.5; Separate homework HW4.5. Constructed teaching example.

Measure the Training Gain From a Split

Within-node variation uses squared distances from the group mean. Impurity here is their average.

1. One group: mean 46; training squared error 3360.
2. Cut at 50: two squared-error sums of 80.
3. Choose total 160, a training reduction of 3200.



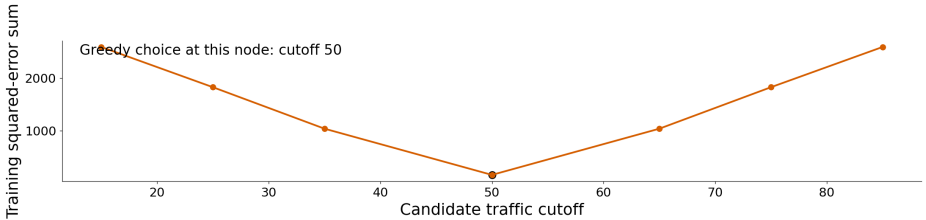


Lecture code 4.6; Separate homework HW4.6. Constructed teaching example.

Greedy Fitting Chooses the Best Current Step

Greedy fitting chooses the best current training step, not an entire future tree.

1. One group: mean 46; training squared error 3360.
2. Cut at 50: two squared-error sums of 80.
3. Choose total 160, a training reduction of 3200.

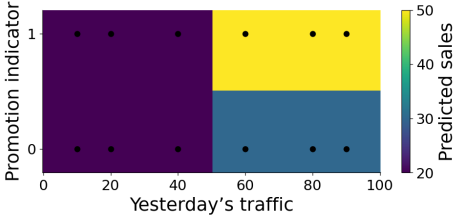


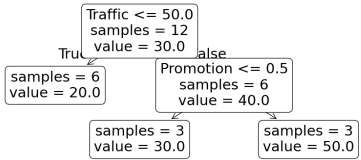
Lecture code 4.6; Separate homework HW4.6. Constructed teaching example.

Questions Divide Input Space

Input space uses input coordinates. A partition divides it; an axis-aligned split asks about one coordinate.

1. Build 12 traffic/promotion pairs with outcomes 20/30/50.
2. Split on traffic, then promotion within high traffic.
3. Check the four predictions: 20, 20, 30, 50.



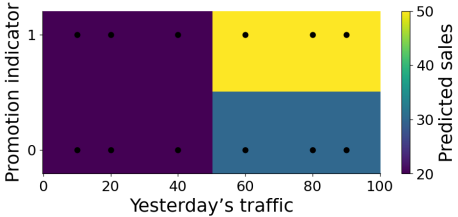


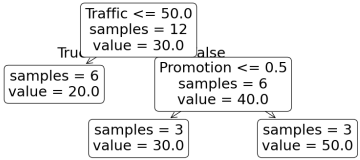
Lecture code 4.7; Separate homework HW4.7. Constructed teaching example.

A Tree Can Represent an Interaction

Asking about promotion only after the high-traffic answer represents an interaction.

1. Build 12 traffic/promotion pairs with outcomes 20/30/50.
2. Split on traffic, then promotion within high traffic.
3. Check the four predictions: 20, 20, 30, 50.



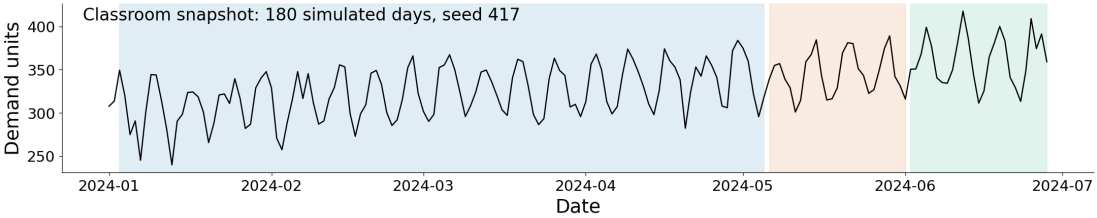


Lecture code 4.7; Separate homework HW4.7. Constructed teaching example.

The Classroom Dataset and Its Information Boundary

Simulation creates artificial observations from stated rules; these are not market data.

1. Load the 180-day local classroom simulation.
2. Create two earlier-demand inputs; keep 178 usable rows.
3. Split by date into 124 / 27 / 27 cases.

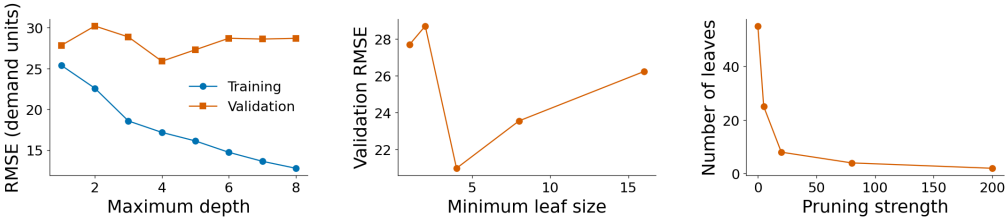


Lecture code 4.8; Separate homework HW4.8. Constructed teaching example.

Regularize a Tree by Restricting Its Detail

Minimum leaf size limits final-group size; pruning removes branches from a fitted tree.

1. Depth panel: hold minimum leaf size at 2.
2. Leaf-size panel: hold maximum depth at 8.
3. Pruning panel: count leaves as pruning strength increases.



Lecture code 4.9; Separate homework HW4.9. Classroom simulation.

Choose Depth and Leaf Size Together

A grid tries a list of candidate settings; here every depth/leaf-size pair is compared.

1. Try 8 depths $\times$ 5 leaf sizes $=$ 40 pairs.
2. Use the same training and validation rows for every pair.
3. Select depth 5, leaf size 4; validation RMSE 20.63.

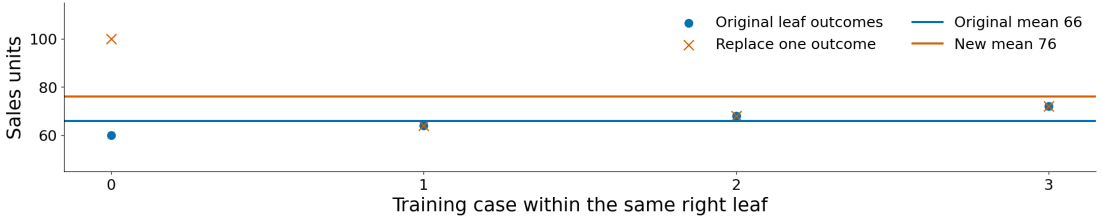


Lecture code 4.10; Separate homework HW4.10. Classroom simulation.

Keep the Prediction Case Fixed When Checking Stability

A model-sensitivity comparison keeps the prediction case fixed.

1. Hold the split and input 75 fixed.
2. Replace outcome 60 with 100 in the same leaf.
3. Compare the means: $66 \rightarrow 76$.

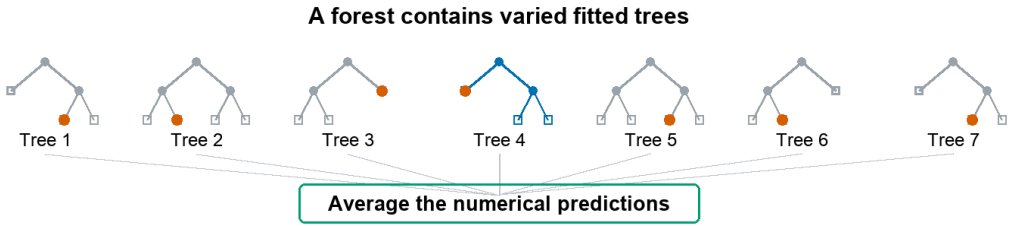


Lecture code 4.11; Separate homework HW4.11. Constructed teaching example.

Why the Name Random Forest?

An ensemble combines fitted models; a random forest averages varied fitted trees.

1. One diagram represents one fitted tree.
2. Expand to a collection and route the same case through each.
3. Average numerical predictions from the member trees.

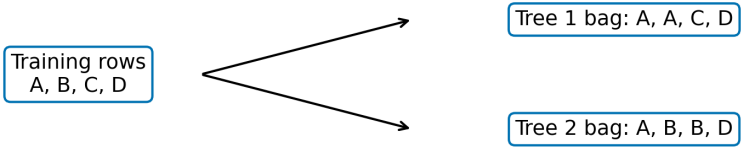


Lecture code 4.12; Separate homework HW4.12. Constructed teaching example.

Bootstrap Rows Within the Training History

With replacement means a selected row remains eligible to be drawn again.

1. Start each tree from the original pool A,B,C,D.
2. Draw A,A,C,D for one and A,B,B,D for another.
3. Identify repeated and missing rows in each bag.

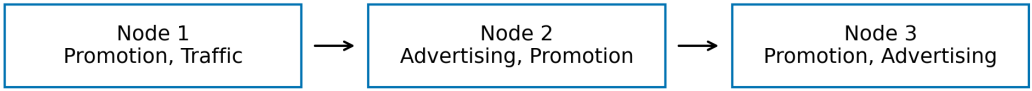


Lecture code 4.13; Separate homework HW4.13. Constructed teaching example.

Randomize Candidate Inputs and Record the Seed

Feature subsampling restricts the input columns that may compete at a split.

1. Start with four named input columns.
2. Draw two candidates per node with seed 417.
3. A subset may repeat; a seed makes the draws reproducible.



Lecture code 4.14; Separate homework HW4.14. Constructed teaching example.

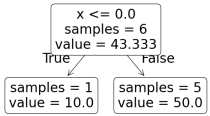
Read the Forest Average

$$\hat{y}_{\text{forest}}(x) = \frac{1}{B}\sum_{b=1}^{B}\hat{f}_b(x).$$

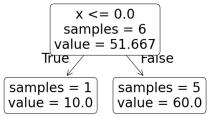
x: one case; $\hat{f}_b(x)$: tree b's prediction; B: tree count.

1. Fit three stated bootstrap bags from the same six-row pool.
2. At x=1, read the fitted leaf means 50, 60, 70.
3. Average the three predictions to obtain 60.

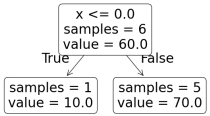
Tree 1: x=1 predicts 50



Tree 2: x=1 predicts 60



Tree 3: x=1 predicts 70



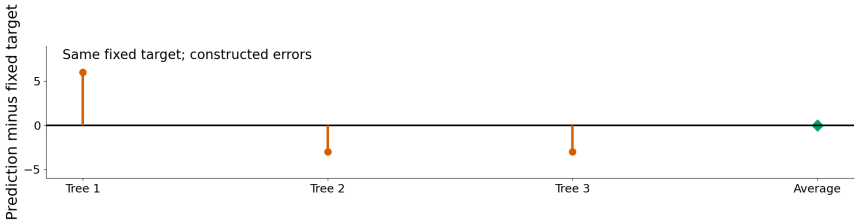
Same case x = 1: (50 + 60 + 70) / 3 = 60

Lecture code 4.15; Separate homework HW4.15. Constructed teaching example.

Different Errors Can Offset in an Average

Compare individual prediction errors with their average, holding the target case fixed.

1. Use one fixed target and errors +6, -3, -3.
2. Their mean is 0: the misses offset.
3. Repeat with +6,+6,+6; the mean remains +6.



Lecture code 4.16; Separate homework HW4.16. Constructed teaching example.

A Shared Error Survives Averaging

Error correlation describes how strongly tree errors move together. A shared error survives averaging.

1. Use the same target and errors $+6$, $+6$, $+6$.
2. Calculate their mean: $+6$.
3. A shared error survives averaging.



Lecture code 4.16; Separate homework HW4.16. Constructed teaching example.

The Forest Still Obeys the Time Boundary

Random tree construction does not authorize future rows in the training history.

1. Fit and resample within the 124-row training history.
2. Use the next 27 rows for model development.
3. Keep the final 27 rows for settled procedures.

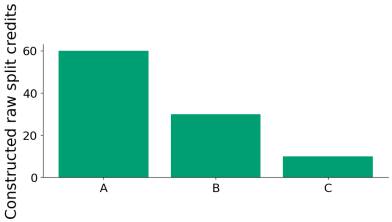


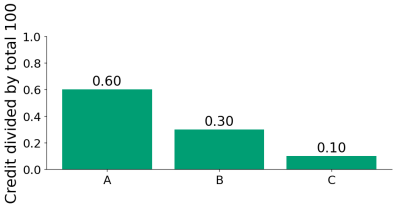
Lecture code 4.17; Separate homework HW4.17. Constructed teaching example.

Calculate Split-Credit Shares by Hand

Normalizing divides each credit by a positive total so the shares sum to one.

1. Add constructed credits $60 + 30 + 10 = 100$.
2. Divide each credit by 100: 0.60, 0.30, 0.10.
3. Interpret shares of split credits, not future accuracy.





Lecture code 4.18; Separate homework HW4.18. Constructed teaching example.

Importance Is a Score With a Definition

Feature importance measures input use under a specified scoring procedure.

1. Read the fitted two-input classroom forest.
2. Its split-credit shares are approximately 0.699 and 0.301.
3. Keep this training score distinct from a shuffle diagnostic.

Training split importance

Credit for separating training outcomes

Permutation importance

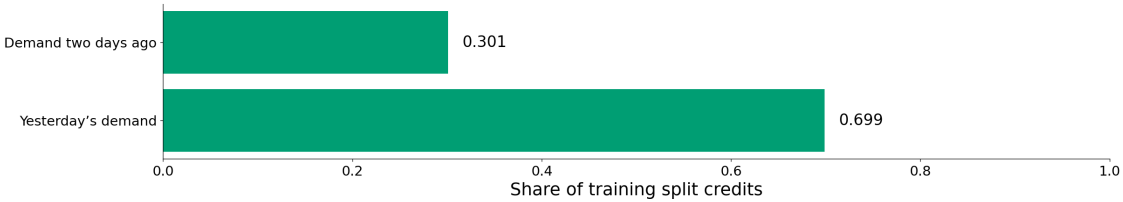
Evaluation error change after rearranging an input

Lecture code 4.19; Separate homework HW4.19. Constructed teaching example.

Training Split Importance Is a Share of Credits

An importance number has meaning only under its stated scoring procedure.

1. Read the fitted two-input classroom forest.
2. Its split-credit shares are approximately 0.699 and 0.301.
3. Keep this training score distinct from a shuffle diagnostic.



Lecture code 4.19; Separate homework HW4.19. Classroom simulation.

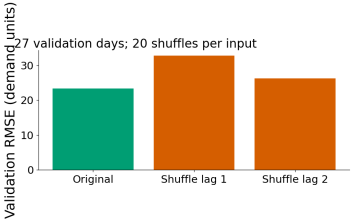
Permutation Importance Breaks One Column's Alignment

Permutation rearranges entries without changing their values; then reevaluate the fixed model.

1. Keep model, outcomes and 27 validation dates fixed.
2. Shuffle one column twenty times.
3. Compare RMSE 23.33 with means 32.72 and 26.27.

Five-row illustration of rearrangement

Day 1: 318.0	354.8
Day 2: 338.7	318.0
Day 3: 354.8	339.1
Day 4: 356.9	338.7
Day 5: 339.1	356.9
Original column	Rearranged

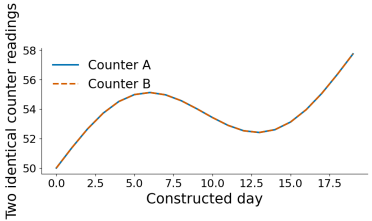


Lecture code 4.20; Separate homework HW4.20. Classroom simulation.

Related Inputs Can Hide Each Other's Importance

Related predictors can supply substitute information to a fitted model.

1. Create two identical constructed counter traces.
2. Rearrange one; the second trace remains intact.
3. Distinguish the diagram from a measured importance score.



Shuffle A's rows

B still carries the same signal

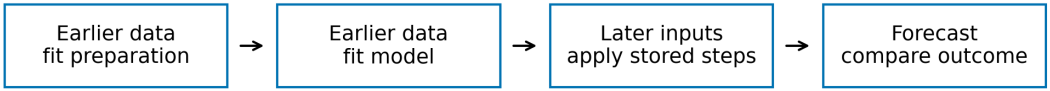
A small score need not mean no information

Lecture code 4.21; Separate homework HW4.21. Constructed teaching example.

Compare the Whole Prediction Pipeline

A pipeline applies preparation and fitting in order, then uses the stored steps for later inputs.

1. Fit any learned preparation on earlier training data.
2. Fit the model using the prepared training inputs.
3. Apply the stored steps to later inputs and evaluate.

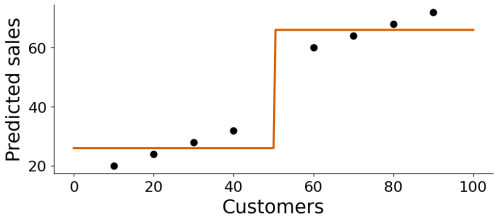


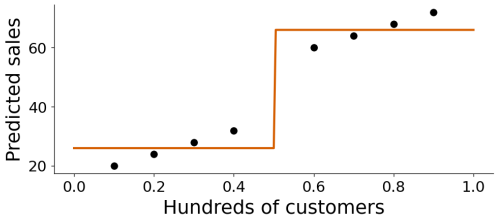
Lecture code 4.22; Separate homework HW4.22. Constructed teaching example.

Change Input Units, Preserve the Tree Groups

Changing reporting units preserves these groups when larger input values remain larger.

1. Fit the same eight pairs in customers and hundreds.
2. Compare cutoffs 50 and approximately 0.5.
3. Queries 35/75 and 0.35/0.75 both predict 26/66.



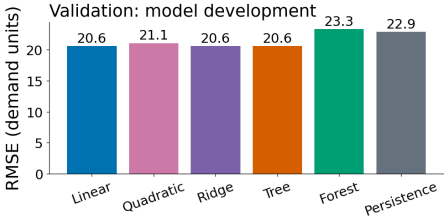


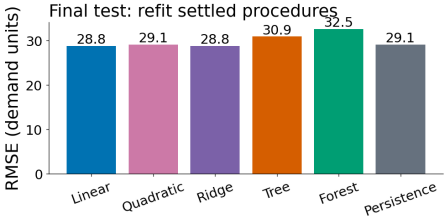
Lecture code 4.22; Separate homework HW4.22. Constructed teaching example.

Choose From Measured Error and Operating Cost

Model comparison uses the same target, dates and units, with settings chosen before final testing.

1. Compare all candidates on identical validation dates.
2. Refit settled procedures on training plus validation.
3. Report test results without selecting from the test.



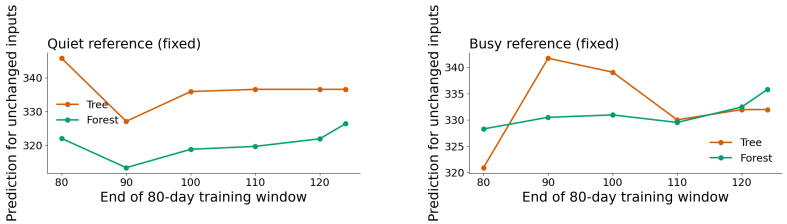


Lecture code 4.23; Separate homework HW4.23. Classroom simulation.

Separate Refit Sensitivity From Changing Cases

Changing the training sample and changing the prediction case answer different questions.

1. Choose two fixed reference input vectors.
2. Refit six 80-day windows inside training history.
3. Compare predictions within each unchanged-case panel.

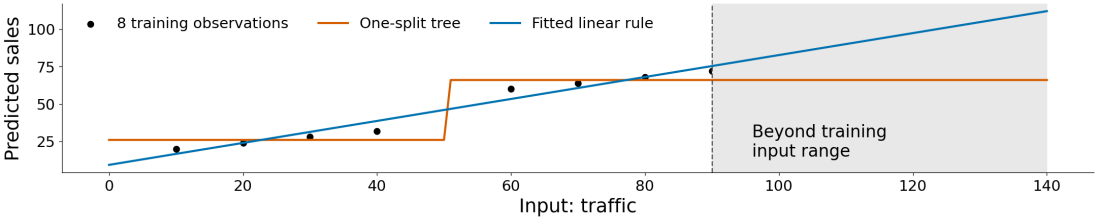


Lecture code 4.24; Separate homework HW4.24. Classroom simulation.

A Forest Does Not Continue a Trend Automatically

Extrapolation applies a rule outside represented input conditions.

1. Fit the same eight pairs, with largest input 90.
2. Extend queries to 140 without adding outcomes.
3. The tree stays at leaf mean 66 outside its range.



Lecture code 4.25; Separate homework HW4.25. Constructed teaching example.

Classroom Examples and Separate Coding Homework

- Classroom notebook: complete worked examples 4.1–4.25.
- Separate student homework: HW4.1–HW4.25, with provided code and marked blanks.
- Each task cites the matching lecture-code section.
- Run the assigned tasks and submit the completed notebook with outputs.

Explain the Rule Before Recommending It

- Trace one case through a tree and compute its leaf mean.
- Explain averaging, diversity and shared error.
- State the validation result and the importance procedure.
- Lecture 5: limits of the rule and the cost of a decision.